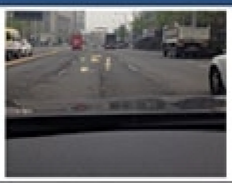
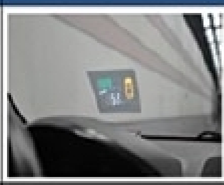
are growing at a steady rate. We are trying to reach our audience currently through partners and other sources like social media marketing. Our focus will be on water treatment and water re-use, and we are working on new innovations that can treat waste water consistently."

—Ayushee Sharma

2 HOLOGRAMS FOR DRIVER NAVIGATION

Once photography was considered to be the best means of recording visual information, but with development of holography, the perception about 2D images changed. Holography is a technique that creates a 3D photograph known as a hologram. A hologram has three dimensions: height, width and depth, whereas a photograph shows only height and width.

Division	BMW CAR HUD (car standard)	TSA Tuning Car HUD (tuning car standard)	MAIN INFOSYSTEM (tuning car standard)
			
Price	US$ 3500	US$ 880	US$ 44
Colour	1~3 colours	3~5 colours	3 colours
Function	1. Use reflective film 2. Using GPS vehicle navigation 3. Separate software using purchasing costs	1. Use reflective film 2. Using GPS vehicle navigation 3. Separate software using purchasing costs	1.The use of one of its film development 2.The mobile navigation shock sensor technology 3. Separate use of software with no T-MAP

Holography records the image of a 3D object on a film so that upon reconstruction, or playback, the constructed image of the object is 3D. It enables the viewer to see the front and sides of an object visible in the hologram by moving from side to side.

Holograms are made by bouncing laser light off a photographic film. When the wave of light hits an object, it takes the shape of the object. It is similar to imprinting a shape on a piece of modelling clay by pressing it against that object. The photographic film is like the film used in a conventional camera but can capture even the tiniest details. It can capture images by recording where laser light hits it and create a hologram.

One cannot do anything when exhausted from tedious driving. This driving may cause accidents and/or violation(s) of traffic regulations. To prevent those situations, Holo navigation has been designed for smartphones and cars with a touch system. It can effectively be used during both day and night.

"This product completes the fusion and convergence of mobile and automobiles through motion sensors and hologram technology. We plan to upgrade navigation 2.0 that has motion sensors to navigation 3.0 that features wireless charging. With this technology, future driving will be safe, fun and convenient," says Park Icy Hyun, chief executive officer, Main Information System Co. Ltd.

Hologram navigation can solve the coherence of sunlight, realise three colours and reduce eye fatigue by using HUD grade holograms. Since the navigation program screen included in the mobile device is projected onto the final product through hologram technology, the final price also decreases.

To actively reflect the cognition of the driver in the emergency/dangerous zone, it is important to harmonise the motions of the driver with the automobile. The driver can get involved more actively when driving and stability can be increased through constructing fused and converged devices using motion technology. Motion sensor technology lets the driver, automobile and external cars communicate with each other at the same time.

Holograms for children's content play devices are also available as a patented technology for mobile convergence solutions. Holographic displays used for advertising is another important application of this technology, which can be seen commonly.

Fusion of various technologies and diverse solutions based on transparent OLEDs, 4D holograms and 5G communications are responsible for rapid changes in display technology in automobiles. Connected car products are appearing on vehicles as vehicle software platforms. To cope with rapid changes in the automobile industry one needs to move from hardware (manual) to software (automated) platforms.

—Deepshikha Shukla

3 DRIVE TO THE SKY WITH SKYDRIVE

Mobility has expanded people's ability to move and enriched their lives by providing them with the joy of going farther and manipulating their own way. However, roads are necessary for cars, buses and trains, and airports and runways are required for aeroplanes. But infrastructure maintenance requires a large investment. In emerging countries where population explosion is expected in the future, a lack of transportation infrastructure may hinder growth.

Hence, to realise the true freedom of movement, compact flying cars with vertical takeoff and landing technology can be a boon, as these will not need roads and/or runways to lift off. In addition, it is necessary to ensure